



Cracking and vertical preferential flow through landfill clay liners

J.H. Li ^{a,b,*}, L. Li ^a, R. Chen ^a, D.Q. Li ^c

^a Department of Civil and Environmental Engineering, Harbin Institute of Technology Shenzhen Graduate School, Shenzhen, 518055, People's Republic of China

^b Centre for Offshore Foundation Systems, The University of Western Australia, Crawley, WA, 6009, Australia

^c State Key Laboratory of Water Resources and Hydropower Engineering Science, Wuhan University, 8 Donghu South Road, Wuhan, 430072, People's Republic of China

ARTICLE INFO

Article history:

Received 29 April 2015

Received in revised form 17 August 2015

Accepted 19 March 2016

Available online 25 March 2016

Keywords:

Landfill

Clay liner

Crack

Infiltration

Root

Unsaturated soil

ABSTRACT

Cracking is a major concern in the long-term performance of landfill clay liners. Cracking of clay liners is affected by the roots of the vegetation planted to control erosion. While vegetation roots may restrict cracking they may introduce preferential flow through the liners. The impact of vegetation roots on the hydraulic properties of clay liners remains unclear. This study investigates the effect of vegetation roots on crack formation and preferential flow through clay liners. Three experimental soil columns were used to observe crack formation by exposure to the weather and to measure the flow through them caused by artificial rainfall. The clay in each column was compacted to 90% relative compaction. The first column was not vegetated and no crack developed in it. The second column was vegetated and underwent drying–wetting cycles outdoor. The third column did not have vegetation but underwent drying–wetting cycles outdoor. Cracks developed in the second and third columns due to the exposure to the weather. Image analysis in the experimental soil columns revealed that vegetation roots restricted crack formation, especially in the early stages of drying–wetting cycles. Infiltration tests performed on the experimental soil columns showed that the unsaturated hydraulic conductivity of the vegetated soil in the second column was respectively about two orders and one order of magnitude larger than those of the soils in the first and third columns. The experimental results imply that the effect of vegetation roots on the preferential flow through landfill clay liners should be considered in practical designs.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Landfill clay liners, consisting of low-permeability materials (e.g. clays), are one of the most important means to minimize water infiltration, gas migration, and leachate generation in landfills. However, the clay liners are prone to cracking during long-term drying–wetting and/or freezing–thawing cycles, which compromises the integrity of the clay liners (Andersland and Al-Moussawi, 1987; Albright et al., 2006; Li and Zhang, 2010; Costa et al., 2013). Desiccation cracks in barrier systems have been a major concern in the UK (Hewitt and Philip, 1999). Desiccation cracks with widths exceeding 10 mm and depths up to 30 mm have been observed by Miller and Mishra (1989). The desiccation cracks become a severe problem that affects the long-term performance. Cracks with widths of 13 mm and maximum crack depths of 1000 mm were observed in a Wisconsin landfill site after three years of exposure. Cracks can significantly accelerate water infiltration into the clay liners, and in turn, increase the leachate generation (Snow, 1969; Yuen et al., 1998; Rayhani et al., 2008; Li et al., 2009b, 2011).

The degradation of clay liners can be relieved by restricting cracking in clays. Soil clay liners for landfills are often vegetated after closure in order to minimize surface erosion (Vaugh et al., 1994). Vegetation roots have been reported as a soil reinforcement, which can provide additional shear strength to soils (Wu and Watson, 1998; Roering et al., 2003). Zhou et al. (2009) found that the root of *Robinia pseudoacacia* could resist soil cracking by performing like a fiber in soil. Sinnathamby et al. (2014) reported that root growth into desiccation cracks hindered self-healing of cracks. It is also reported that water can infiltrate along grass roots bypassing the soil matrix around (Mitchell et al., 1995). Devitt and Smith (2002) found that dead root systems formed macro pores and increased the downward water flux in desert soils. Hence it is arguable whether grass roots restrict soil cracking and hence hinder water infiltration through cracks or not.

As a positive factor the vegetation roots have the potential to restrict the cracking in clay liners. As a negative factor the roots may introduce preferential flows into the clay liners. The impacts of vegetation roots on the landfill clay liner are controversial and need further investigation (Zhu and Zhang, 2015).

This paper investigates the restrained cracking behavior and the preferential flow imposed by vegetation roots in clay liners in order to provide new evidence to the controversial observations in the field. Three clay liners (i.e. compacted soil, vegetated soil with cracks and cracked soil without vegetation) were carefully prepared in the soil

* Corresponding author at: Department of Civil and Environmental Engineering, Harbin Institute of Technology Shenzhen Graduate School, Shenzhen, 518055, People's Republic of China.

E-mail address: celijinhui@gmail.com (J.H. Li).

columns. The cracking behavior in these soil columns was observed and quantified first, followed by infiltration tests through these columns. The cracking pattern and intensity as well as the hydraulic conductivity of the three clay liners were obtained, which sheds light on the vertical preferential flow of the clay liners. The evapotranspiration caused by the vegetation is out of the scope of this paper.

2. Apparatus

The experimental setup consisted of a cylindrical soil column that was equipped with tensiometers and moisture content sensors, a rainfall simulation system, and a water collection system (see Fig. 1). The rainfall simulation system consisted of a cylindrical rain simulator (50 mm in height and 200 mm in diameter) connecting to a Mariotte bottle. The bottom of the rain simulator was a porous disk with 41 small holes 0.2 mm in diameter, which simulates rainfall dropping from the rain simulator (Fig. 2). Water in the rain simulator was provided by the Mariotte bottle, which maintains a steady water pressure and thus a steady rainfall intensity. A water collection system collected the overflow from the soil surface and the water leaching from the bottom of the soil column. A water container and an electronic balance were used to collect and measure the mass of the leaching water and runoff water, respectively (see Fig. 1).

The soils were compacted to 90% relative compaction in a transparent cylindrical column (700 mm in height and 200 mm in diameter). A porous stone disk with a slope of 5% was installed at the center of the column base for better drainage and preventing channel clogging. Tensiometers and moisture content sensors were installed along the column at different heights as shown in Fig. 1. The tensiometers monitored the pore water pressure changes along the column. The suction

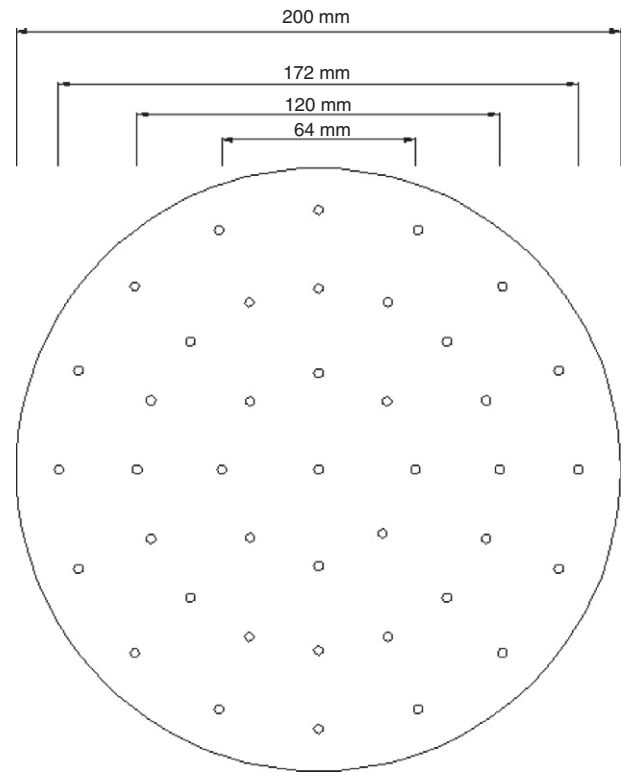


Fig. 2. Dimensions of the bottom of the rain simulator (not in scale).

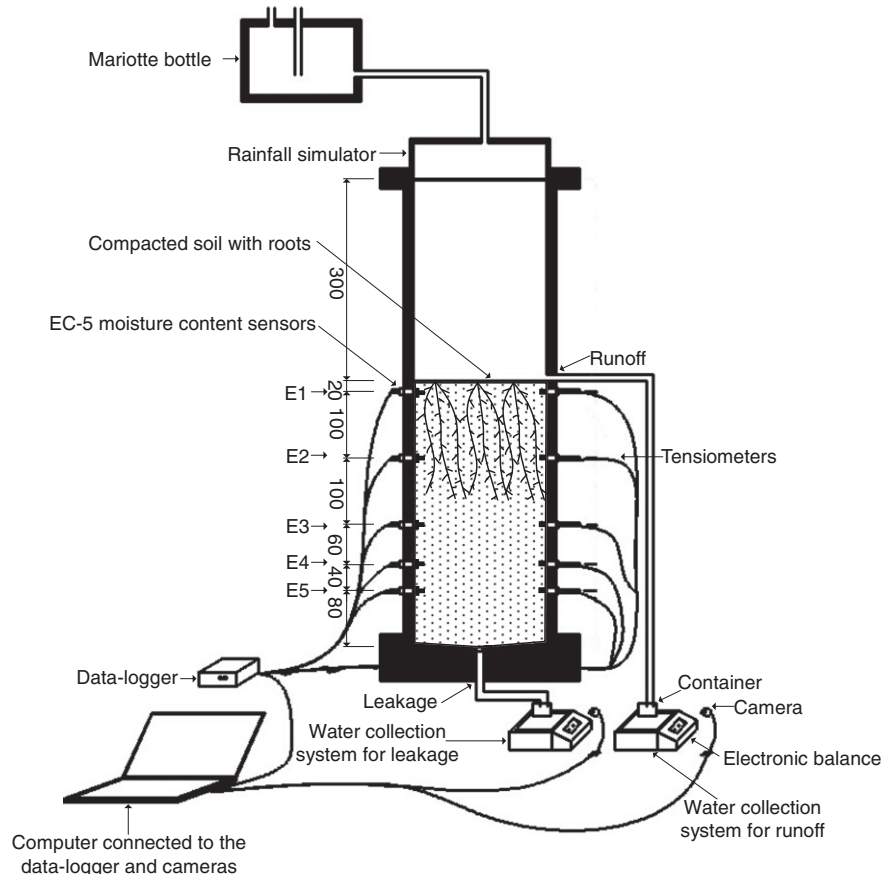


Fig. 1. Sketch of the soil column system (unit: mm).

range of the tensiometers was from 0 kPa to 90 kPa. The accuracy of the tensiometers was ± 1.5 kPa. The moisture content sensors monitored the water content changes in the soils, with a measuring range of 0–50% and an accuracy of 1%–2%. Both the tensiometers and the moisture content sensors were connected to a data-logger, which continuously records pore water pressure and water content readings.

3. Materials

A residual soil taken from an excavation site in Hebei Province, China, was used in the experiments. The soil properties are summarized in Table 1. The particle size distribution was measured according to BS1377 (1990) and is shown in Fig. 3. The soil is classified as inorganic clay of low plasticity (ML) according to the Unified Soil Classification System (USCS).

The seeds of Tall Fescue were sown in the soil column to investigate the effect of vegetation roots. Tall Fescue is a perennial grass that is commonly used for soil conservation around the world (Stuedemann and Hoveland, 1988; Spyreas et al., 2001; Rahman and Saiga, 2005). It covers a large area of the eastern USA, and is increasingly popular in Africa, Australia, China, New Zealand, and South America (Rudgers and Clay, 2007). While growing, its main roots develop downwards into the soils. Lateral roots grow at nearly equal spacing along the main roots (Fig. 4).

4. Experimental tests

Both image analysis and infiltration tests were conducted to investigate the influence of cracks and roots on the vertical flow through clay liners. The first soil column consisted of compacted soils without grass and cracks. This soil column represents a soil condition immediately after being compacted in the field. The second soil column was firstly compacted and then sowed with Tall Fescue seeds. This soil column was put outdoor and exposed to the elements of weather (e.g. insolation, wind, temperature, humidity) for six weeks. During this period, the seeds sprouted and grew well in the column. In the meantime, cracks developed gradually in the soil. Soils in the third column were compacted as the first and second columns were. After compaction, this column experienced exactly the same drying–wetting cycles as the second one did. However, no grass was planted in the third column. Hence, cracks developed without the restraints from roots in the third column. The conditions of the three soil column tests are listed in Table 2.

4.1. Soil sample preparation

- (1) The first soil column. The soil was first prepared to the initial water content of 0.15 and then compacted to 90% relative compaction uniformly according to ASTM D698-12 (2012). Vaseline was applied along the inner-wall of the column before compaction to avoid preferential flow along the wall in the subsequent

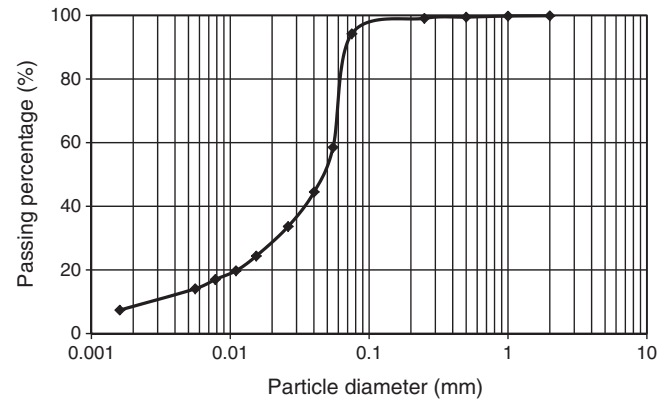


Fig. 3. Particle size distribution.

infiltration test. This soil column was neither vegetated nor exposed to the elements of weather. No cracks developed in this soil column.

- (2) The second soil column. The compaction process of the second soil column was the same as that of the first one. After compaction, 50 g/m² of grass seeds were distributed evenly on the soil surface. Then the seeds were covered by a thin layer of soil to promote growing of the grass. This vegetated soil column was then placed outdoor. Natural drying processes through solar radiation, heating and wind as well as wetting processes through irrigation every two days were applied on the soil column. After six weeks of drying–wetting cycles, Tall Fescue had grown very well. Cracks also developed in the vegetated soils.
- (3) The third soil column. Similar compaction process was applied to the third soil column. After compaction, this soil column was placed outdoor and exposed to the elements of weather at the same time as the second column. The soil experienced exactly the same drying–wetting cycles as the second one did. Therefore, the only difference between the second soil column and the third one was that no vegetation was planted in the third one. During the six weeks of drying–wetting cycles, cracks developed freely in this soil column without the influence of vegetation.

4.2. Experimental procedure

The experiments were divided into two stages: (a) crack image analysis and (b) infiltration test. A digital imaging method (Li and Zhang, 2010) was used to log the crack development during the six weeks of drying–wetting cycles. Photographs were taken every two days on the soil surface with the camera fixing at the same place and height to ensure the quality. The pixel of the camera was 4416 × 3312. The accuracy of each photograph can be determined by

Table 1
Basic properties of the clay.

Property	Value
Maximum dry density, ρ_d , (kg/m ³)	1710
Optimal water content, w_{opt}	0.19
Liquid limit, w_L	42.19
Plastic limit, w_p	19.65
Plastic index, I_p	22.54
Coefficient of uniformity, C_u	15
Coefficient of curvature, C_c	2.6

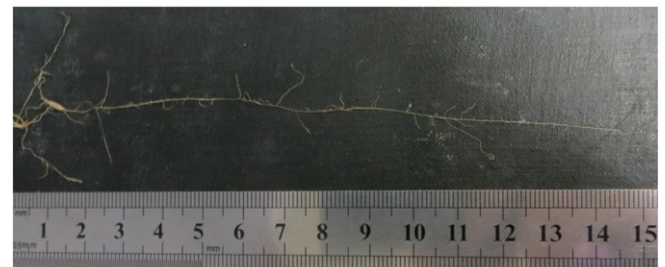


Fig. 4. The roots of Tall Fescue.

Table 2
Details of the three soil column tests.

Test	Conditions	Crack development	Grass roots	Degree of compaction	Initial water content
I	Without vegetation and indoor exposure	No	No	90%	0.15
II	With vegetation and outdoor exposure	Yes	Yes	90%	0.19
III	Without vegetation and outdoor exposure	Yes	No	90%	0.21

the ratio between the size of the photograph and the pixels, which was about 0.06 mm. On the 15th, 27th and 37th days, the grass was cut, and the soil surface and the cracks were clearly exposed. The cracking

behavior in the vegetated soils and in the bare soils could then be compared. The depth of roots could be observed through the transparent column in the second soil column.

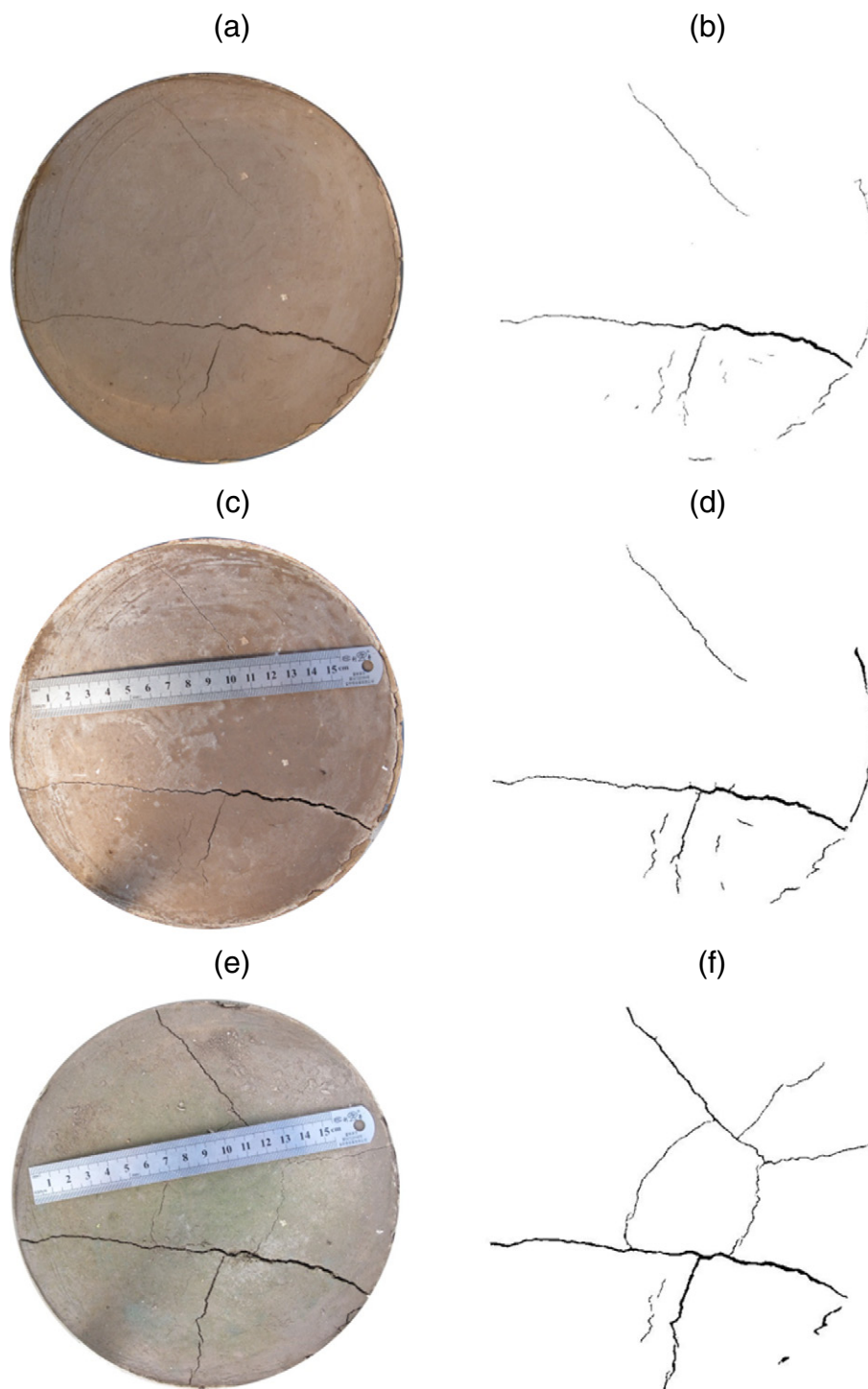


Fig. 5. Photographs of surface cracking on the third soil column at different times: (a) 15 days; (b) the binary image of Fig. 5a; (c) 27 days; (d) the binary image of Fig. 5c; (e) 37 days; (f) the binary image of Fig. 5e.

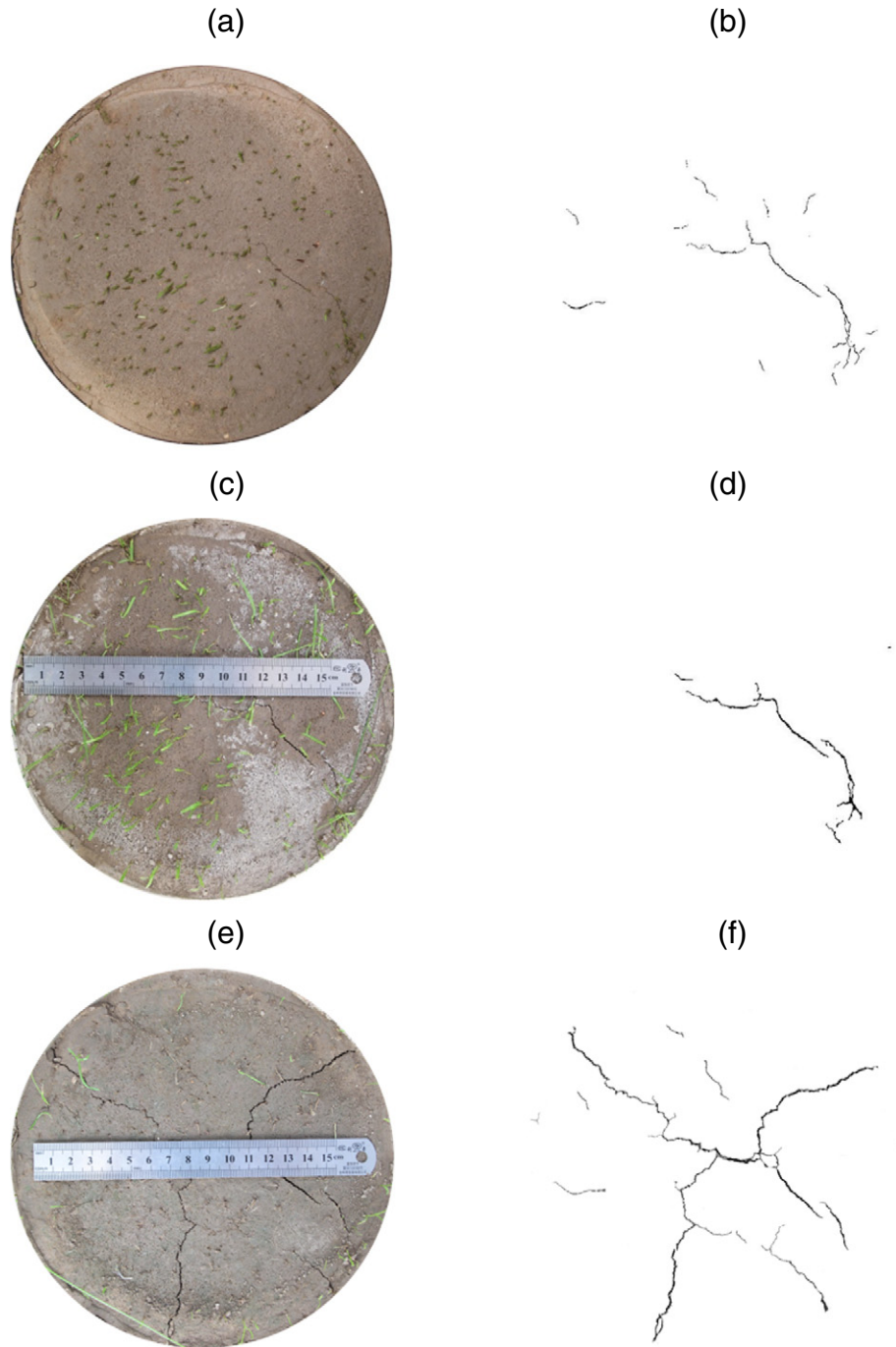


Fig. 6. Photographs of surface cracking on the second soil column at different times: (a) 15 days; (b) the binary image of Fig. 6a; (c) 27 days; (d) the binary image of Fig. 6c; (e) 37 days; (f) the binary image of Fig. 6e.

For the first soil column, the infiltration test was conducted immediately after compaction. For the second and third soil columns, infiltration tests were conducted after the crack image analysis. As soon as the image analysis was completed, the rain simulator was installed above the column. Then five tensiometers and five moisture content sensors were installed along the soil column. These sensors were connected to a data-logger which collected the data every one second. Two water containers were used to collect the water from runoff and leakage separately. The water mass changes in the two containers were recorded through a camera and a computer.

Once the experimental system had been set up a steady rain with an intensity of 8.64×10^{-6} m/s was applied to simulate a rainstorm. During this process, the changes in suction and water content along

the soil column were recorded and used to investigate the infiltration rate as well as the unsaturated hydraulic conductivity.

5. Data interpretation

The methods to interpret the cracking behavior and the hydraulic properties of the clay liners were presented in this section.

5.1. Crack intensity factor

A crack intensity factor (*CIF*) is often used to quantify the cracking on the soil surface, which is defined as the ratio between the crack area and

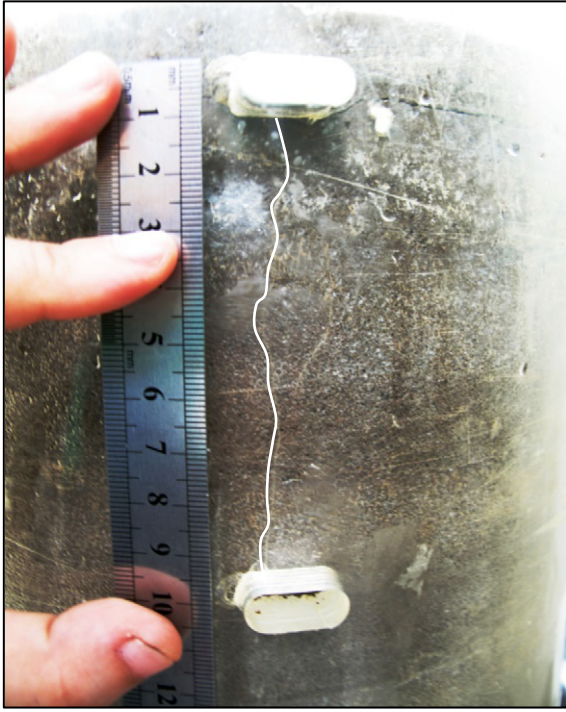


Fig. 7. The depth of grass roots.

the entire soil surface area (Miller et al., 1998; Yesiller et al., 2000),

$$CIF = \frac{A_c}{A_s} \quad (1)$$

where A_c is the area of cracks on the soil surface, and A_s is the total area of the soil surface. To calculate the areas the photographs were transformed into binary images. The pixels that are occupied by the cracks and by the soil surface in a binary image are counted respectively. In this way the aperture values of the cracks can also be obtained from the photographs.

5.2. Infiltration rate and hydraulic conductivity

The steady-state, instantaneous profile, and wetting front advancing methods are often used to measure the unsaturated hydraulic conductivity of unsaturated soils (Fredlund and Rahardjo, 1993; Lu and William, 2004; Li et al., 2009a; Puppala et al., 2013; Romero, 2013). In this study the instantaneous profile method is adopted.

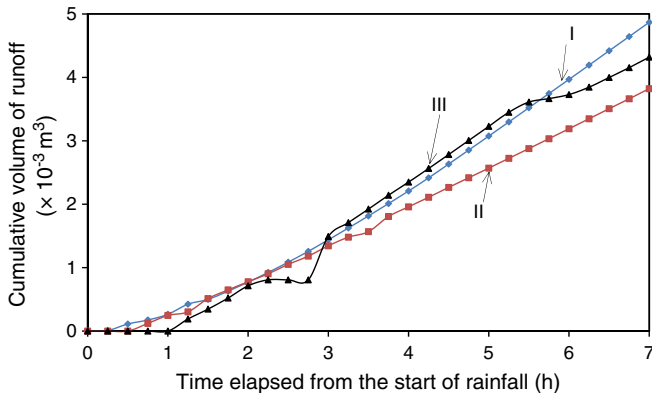


Fig. 8. Cumulative volume of runoff from the three soil columns.

Table 3

Leakage from the three soil columns.

Properties	Test I	Test II	Test III
Leakage start time (hour)	41	3.05	20
Cumulative volume of rainfall ($\times 10^{-3} \text{ m}^3$)	31	4.4	15
Cumulative volume of leakage ($\times 10^{-3} \text{ m}^3$)	0.09	0.56	0.07
Percentage of leakage (%)	0.3	12.7	0.5
Rate of leakage (m/s)	5.9×10^{-8}	3.0×10^{-7}	1.0×10^{-7}

As rainfall started, water flowed downwards and induced increasing water content in the soil. The increase in water volume per unit time is defined as the infiltration rate,

$$v = \frac{dV_w}{A_s dt} \quad (2)$$

where v is the infiltration rate and dV_w is the change in water volume during time dt . dV_w could be calculated using the measured volumetric water contents from different sensors,

$$dV_w = A_s \left(\int_j^m \theta_w^{t_i + \Delta t}(x) dx - \int_j^m \theta_w^{t_i}(x) dx \right) \quad (3)$$

where dV_w is the total water volume change in the soil between point j and the bottom of the soil column designated as point m ; $\theta_w^{t_i}(x)$ is the volumetric water content profile as a function of distance x for specific time t_i ; $\theta_w^{t_i + \Delta t}(x)$ is the volumetric water content profile for specific time $t_i + \Delta t$.

Different infiltration rates, v , corresponding to different suctions and different hydraulic gradients, could be obtained. The hydraulic conductivity, k , could be calculated by dividing the infiltration rate by the corresponding mean hydraulic gradient, i ,

$$k = \frac{v}{i} \quad (4)$$

The mean hydraulic gradient is obtained by,

$$i = \frac{(h_{i+1,t1} - h_{i,t1}) + (h_{i+1,t2} - h_{i,t2})}{2 \times \Delta z} \quad (5)$$

where $h_{i+1,t1}$ is the hydraulic head at point $i + 1$ at time t_1 ; $h_{i,t1}$ is the hydraulic head at point i at time t_1 ; $h_{i+1,t2}$ is the hydraulic head at point $i + 1$ at time t_2 ($t_2 > t_1$); $h_{i,t2}$ is the hydraulic head at point i at time t_2 ; Δz is the distance between point i and point $i + 1$.

Computations for the infiltration rates and hydraulic conductivities could be repeated for different points and different time. As a result, the infiltration rates and hydraulic conductivities at various suction values during each test could be computed.

6. Results

6.1. Effects of roots on cracking

In this section the cracking behavior was observed and quantified in the second and third soil columns. Fig. 5 shows the crack development in the third soil column after 15, 27 and 37 days of drying–wetting cycles. In Fig. 5a, a main crack develops across the soil surface accompanied by several minor cracks. Fig. 5b shows the binary image of Fig. 5a, from which the CIF can be calculated. The cracks occupy 86,814 pixels and the soils have 8,426,883 pixels. Hence, the CIF on the 15th day is 1.03%. The accuracy of the CIF is only affected by the numbers of pixels at the crack edges that are hard to identify. Considering the large number of pixels that are occupied by the cracks, the influence of the crack edges is limited.

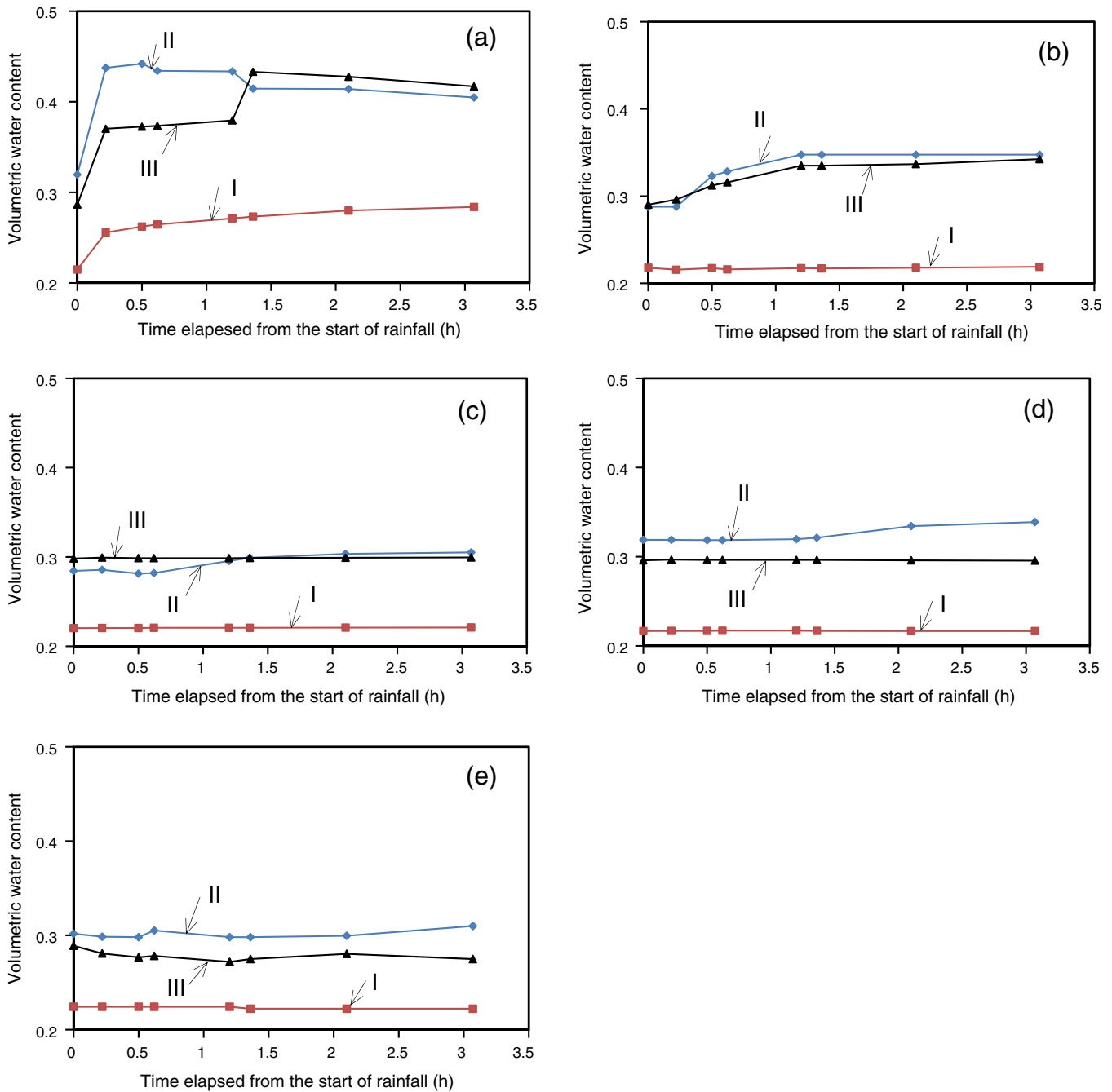


Fig. 9. Volumetric water contents from five moisture content sensors at different times: (a) E1; (b) E2; (c) E3; (d) E4; (e) E5.

As the soils experienced more cycles, cracks continued to develop as shown in Fig. 5c and e. On the 37th day, seven main cracks had developed and connected to each other. These connected cracks formed a typical crack network, which is often observed in the field. Finally the CIF of the third bare soil column increased to 1.89%. The average aperture of the cracks was 0.90 mm.

In the second soil column, cracks developed and roots grew during the drying–wetting cycles. Photographs were taken after cutting the top part of Tall Fescue on the 15th, 27th and 37th days. On the 15th day, several small and tortuous cracks developed in the second soil column (as shown in Fig. 6a). Fig. 6b shows the binary image and its CIF is 0.36%. The CIF of the vegetated soil is much less than that of the bare soil (i.e. 1.03%). It is the vegetation roots that restrict the development of cracks in soils. Generally cracks appear when the tensile stress in the soil is larger than the tensile strength of the soil (Li and Zhang,

2011). The tensile strength of the bare soil was very small, which led to intensive cracks in the third soil column. The tensile strength of the Tall Fescue root was about 49 MPa, which is much larger than that of the soil (Zhang et al., 2002). The high tensile strength of the roots restricted the crack development in the soil. The interface strength (or friction) between the roots and the soils can provide an additional cohesion to restrict cracking (Waldron, 1977; Waldron and Dakessian, 1981). On the 37th day, cracks in the vegetated soils fully developed, but without forming crack networks. The CIF reached 1.81% and the average crack aperture was 0.88 mm. The results show that the vegetation roots restrict the development of cracks, especially in the early stages of drying–wetting cycles. This restriction effect becomes limited as the drying–wetting cycles continue. The roots could be observed through the transparent column wall, which had reached 110 mm after six weeks of growth (Fig. 7).

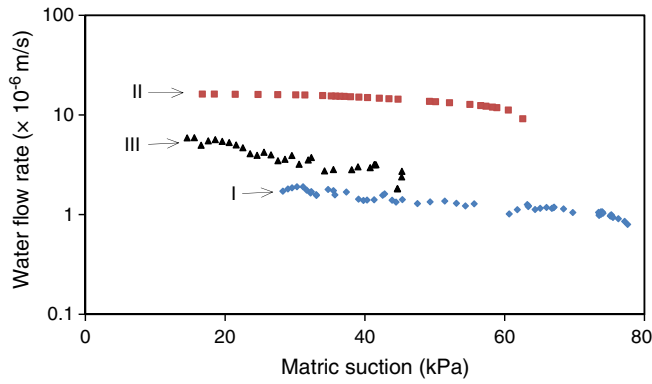


Fig. 10. Water flow rates.

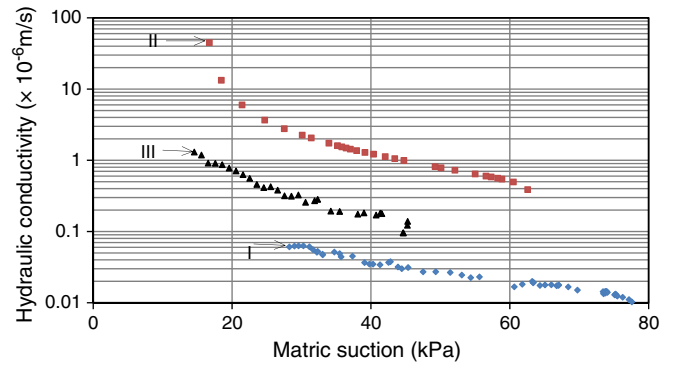


Fig. 12. Hydraulic conductivities.

6.2. Surface runoff

When a heavy storm was applied on the low-permeability clay, a large portion of the rainfall turned into runoff and flowed out through the water collection system (Fig. 1). The cumulative volume of runoff is plotted in Fig. 8. The runoff occurred the earliest in the first soil column where neither cracks nor roots were present. The runoff occurred the latest in the third soil column where cracks were most intensive (i.e. large CIF). This phenomenon indicates that cracked soils absorb more water in the early stage. The runoff in the third soil column increased substantially after 2.75 h of rainfall because the cracks gradually closed during the wetting process.

After 7 h of rainfall, the total amount of runoff in the second soil column (with both cracks and roots) was the least. This indicates that cracks and roots facilitate water infiltration into the soils. When roots were present in the second soil column, there was no sudden increase in runoff, which means that the cracks were still open in the wetting process. This might be attributed to the roots that hindered the self-healing of cracks, and thus led to more water infiltration into the vegetated soil.

6.3. Leakage

As the heavy storm continued, water flowed downwards and finally leaked from the bottom of the soil column. The rainfall was stopped after 4 h of leakage in each column. Leakage was observed after 41 h of rainfall in the first freshly compacted soil column. When cracks developed but no roots were present in the third soil column the leakage started after 20 h of rainfall. The cracks accelerated the percolation in the clay liner. In the second vegetated soil column the leakage occurred after only 3 h of rainfall. The roots in the clay further speeded up the percolation in the liner.

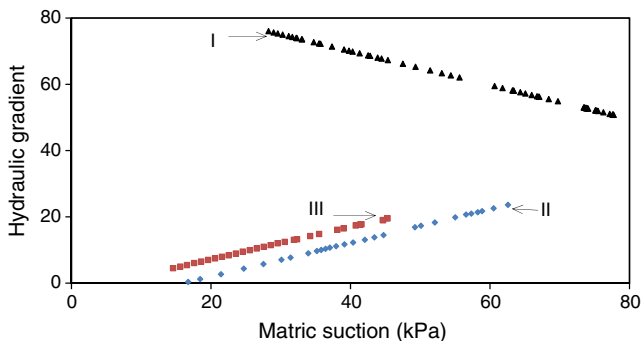


Fig. 11. Average hydraulic gradients.

The ratio between the volume of leakage water and the volume of rainfall is defined as the percentage of leakage (as listed in Table 3). The percentages of leakage are 0.3%, 12.7%, and 0.5% for the first, second, and third soil columns, respectively. The leakage from the third soil column with cracks only is much smaller than that from the second soil column containing both roots and cracks. This indicates that the preferential flow along the roots plays an important role. The rate of leakage can be calculated by dividing the volume of leakage by the duration of leakage. The rates of leakage for the first, second and third soil columns are 5.9×10^{-8} , 3.0×10^{-7} and 1.0×10^{-7} m/s, respectively. The rate of leakage for the soil column with both cracks and roots is the largest. Again, the results show that the performance of a clay liner can deteriorate significantly by the preferential flow along vegetation roots.

6.4. Unsaturated hydraulic conductivity

In this section, the water contents along the soil columns and the flow rates are presented first. The unsaturated hydraulic conductivity can be obtained by combining the water flow rate with the measured pore water pressure using Eq. (4).

The initial water contents of the three soil columns were 0.15, 0.19, and 0.21, respectively. As the second and third soil columns experienced numerous drying–wetting cycles, it was hard to attain exactly the same initial water content for the three soil columns. The changes in volumetric water content at different depths (i.e. E1, E2, E3, E4 and E5 in Fig. 1) are shown in Fig. 9. Fig. 9a shows that sensor E1 (buried at a depth of 20 mm) responds immediately as the rainfall arrives at the soil surface in all of the soil columns. Fig. 9b shows that the water content at 120 mm depth (i.e. sensor E2) increases after 3 h, 0.2 h and 0.5 h of rainfall in the first, second and third soil columns, respectively. The rapid response in the second vegetated soil column further confirms that the vegetation roots promote water infiltration into the soils.

The water flow rates can be calculated through the change in the volumetric water contents using Eq. (2). Fig. 10 shows the water flow rates through the three soil columns. The flow rate in the first soil column increases from 0.8×10^{-6} m/s to 1.9×10^{-6} m/s as the matric suction decreases from 77 kPa to 27 kPa during the rainfall process. When cracks are present in the soils, the flow rate in the third soil column is in the range of 1.8×10^{-6} to 5.8×10^{-6} m/s, which is larger than that in the freshly compacted soil. When both cracks and roots are present in the second soil column, the flow rate increases significantly, which is between 9.2×10^{-6} m/s and 1.6×10^{-5} m/s. Hence, the flow rate in the vegetated soils is about one order of magnitude larger than that in the freshly compacted soil.

The hydraulic gradient can be calculated by the suction measurements from the tensiometers according to Eq. (5). Fig. 11 shows the hydraulic gradient values corresponding to different matric suctions. The hydraulic gradient in the first soil column is much larger than those in the second and the third ones. This is because the suction

differences between two adjacent sensors are large in the freshly compacted soil.

The hydraulic conductivity is calculated using the water flow rates and the hydraulic gradients at different suctions. Fig. 12 shows the variation of the unsaturated hydraulic conductivity with the matric suction for the three soil columns. The hydraulic conductivity decreases as the matric suction increases, which is expected in unsaturated soils (Fredlund et al., 1994; Leong and Rahardjo, 1997; Ng and Leung, 2011; Wang et al., 2013; Cai et al., 2014). When both roots and cracks are present, the hydraulic conductivity ranges from 3.9×10^{-7} m/s to 4.5×10^{-5} m/s. The large variation implies that the preferential effect is more prominent when the soil tends to be saturated. When only cracks are present, the hydraulic conductivity varies from 9.6×10^{-8} m/s to 1.3×10^{-6} m/s, which is about one order of magnitude smaller than that of the vegetated soil. When no cracks and no roots are present, the hydraulic conductivity is between 1.0×10^{-8} m/s and 6.2×10^{-8} m/s, which is two orders of magnitude smaller than that of the vegetated soil.

The saturated hydraulic conductivity increases about three orders of magnitude (from 10^{-8} to 10^{-5} m/s) when both roots and cracks develop in the clay liner. This result is consistent with the field observation of a compacted clay liner in USA where the hydraulic conductivity increased approximate three orders of magnitude over a 4-year service life (Albright et al., 2006). The presence of cracks was then recognized as the key factor of clay liner degradation. This study further identifies that the roots play an important role in the increased hydraulic conductivity. However, the applicability of the current results on the geosynthetic landfill liners needs to be verified as the plant roots have been found to hardly penetrate a geosynthetic liner (Holl, 2002). Hence the results in this study should be limited to the compacted clay liners.

7. Conclusions

The restrained cracking and vertical preferential flow induced by vegetation roots in clay liners were investigated by performing crack image analysis and infiltration tests on three carefully prepared soil columns. The soil compaction, growth of vegetation, crack development and observation, as well as the infiltration tests were all conducted in the soil column, which minimized the disturbance on the soil samples and crack development. The results provide new evidence on the debating effect of vegetation on clay liners.

Results indicate that the vegetation roots are capable of restricting the crack development, especially in the early stages of the drying–wetting cycles. This is because the larger shear strength of rooted systems restricts the cracking in soils. Although the roots restricted the crack development, they hindered the self-healing of cracks during the wetting process, which would increase the water infiltration through the cracks in prolonged rainfall.

The leakage from the vegetated soil column was observed only after 3 h of rainfall and its volume was larger than those in the non-vegetated soil columns. The saturated hydraulic conductivity increased about three orders of magnitude (from 10^{-8} to 10^{-5} m/s) when both roots and cracks developed in the clay liner. The unsaturated hydraulic conductivity of the vegetated soil was about two orders of magnitude larger than that of the freshly compacted soil. These evidences show that the vegetation roots are significant preferential flow pathways and should be considered in engineering design.

Acknowledgments

This research was substantially supported by the Natural Science Foundation of China (Project Nos. 51379053 and 51329901), Rock Mechanics in Hydraulic Structural Engineering (Ministry of Education), as well as Open Research Fund Program of the State Key Laboratory of Water Resources and Hydropower Engineering Science.

References

- Albright, W.H., Benson, C.H., Gee, G.W., Abichou, T., McDonald, E.V., Tyler, S.W., Rock, S.A., 2006. Field performance of a compacted clay landfill final cover at a humid site. *J. Geotech. Geoenviron.* 132 (11), 1393–1403.
- Andersland, O.B., Al-Moussawi, H.M., 1987. Crack formation in soil landfill covers due to thermal contraction. *Waste Manag. Res.* 5 (4), 445–452.
- ASTM D698–12, 2012. Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12 400 Ft-Lbf/Ft³ (600 kN·m/m³)). ASTM International, West Conshohocken, PA.
- BSI (British Standards Institution), 1990. BS 1377–2: Methods of Test for Soils for Civil Engineering Purpose. Part 2: Classification Tests. BSI, Milton Keynes.
- Cai, G.Q., Zhou, A., Sheng, D., 2014. Permeability function for unsaturated soils with different initial densities. *Can. Geotech. J.* 51 (12), 1456–1467.
- Costa, S., Kodikara, J., Shannon, B., 2013. Salient factors controlling desiccation cracking of clay in laboratory experiments. *Géotechnique* 63 (1), 18–29.
- Devitt, D.A., Smith, S.D., 2002. Root channel macropores enhance downward movement of water in a Mojave Desert ecosystem. *J. Arid Environ.* 50 (1), 99–108.
- Fredlund, D.G., Rahardjo, H., 1993. *Soil Mechanics for Unsaturated Soils*. John Wiley & Sons, New York.
- Fredlund, D.G., Xing, A., Huang, S., 1994. Predicting the permeability function for unsaturated soils using the soil–water characteristic curve. *Can. Geotech. J.* 31 (4), 533–546.
- Hewitt, P.J., Philip, L.K., 1999. Problems of clay desiccation in composite lining systems. *Eng. Geol.* 53 (2), 107–113.
- Holl, K.D., 2002. Roots of Chaparral Shrubs fail to penetrate a geosynthetic landfill liner. *Ecol. Restor.* 20 (2), 112–116.
- Leong, E.C., Rahardjo, H., 1997. Permeability functions for unsaturated soils. *J. Geotech. Geoenviron.* 123 (12), 1118–1126.
- Li, J.H., Zhang, L.M., 2010. Geometric parameters and REV of a crack network in soil. *Comput. Geotech.* 37, 466–475.
- Li, J.H., Zhang, L.M., 2011. Study of desiccation crack initiation and development at ground surface. *Eng. Geol.* 123 (4), 347–358.
- Li, X., Zhang, L.M., Fredlund, D.G., 2009a. Wetting front advancing column test for measuring unsaturated hydraulic conductivity. *Can. Geotech. J.* 46 (12), 1431–1445.
- Li, J.H., Zhang, L.M., Wang, Y., Fredlund, D.G., 2009b. Permeability tensor and representative elementary volume of saturated cracked soil. *Can. Geotech. J.* 46, 928–942.
- Li, J.H., Zhang, L.M., Kong, C.P., 2011. Field permeability at shallow depth in a compacted fill. *Geotech. Eng. Proc. ICE* 164 (GE3), 211–221.
- Lu, N., William, J.L., 2004. *Unsaturated Soil Mechanics*. John Wiley & Sons, New Jersey.
- Miller, C.J., Mishra, M., 1989. Modeling of leakage through cracked clay liners – I: state of the art. *J. Am. Water Resour. Assoc.* 25 (3), 551–556.
- Miller, C.J., Mi, H., Yesiller, N., 1998. Experimental analysis of desiccation crack propagation in clay liners. *J. Am. Water Resour. Assoc.* 34 (3), 677–686.
- Mitchell, A.R., Ellsworth, T.R., Meek, B.D., 1995. Effect of root systems on preferential flow in swelling soil. *Commun. Soil Sci. Plant Anal.* 26 (15&16), 1655–2666.
- Ng, C.W.W., Leung, A.K., 2011. Measurements of drying and wetting permeability functions using a new stress-controllable soil column. *J. Geotech. Geoenviron.* 138 (1), 58–68.
- Puppala, A.J., Manosuthikij, T., Chittoori, B., 2013. Swell and shrinkage characterizations of unfilled expansive clays from Texas. *Eng. Geol.* 164, 187–194.
- Rahman, M.H., Saiga, S., 2005. Endophytic fungi (*Neotyphodium coenophialum*) affect the growth and mineral uptake, transport and efficiency ratios in tall fescue (*Festuca arundinacea*). *Plant Soil* 272, 163–171.
- Rayhani, M.H.T., Yanful, E.K., Fakher, A., 2008. Physical modeling of desiccation cracking in plastic soils. *Eng. Geol.* 97 (1), 25–31.
- Roering, J.J., Schmidt, K.M., Stock, J.D., 2003. Shallow landsliding, root reinforcement, and the spatial distribution of trees in the Oregon Coast range. *Can. Geotech. J.* 40 (2), 237–253.
- Romero, E., 2013. A microstructural insight into compacted clayey soils and their hydraulic properties. *Eng. Geol.* 165, 3–19.
- Rudgers, J.A., Clay, K., 2007. Endophyte symbiosis with tall fescue: how strong are the impacts on communities and ecosystems? *Fungal Biol. Rev.* 21, 107–124.
- Sinnathamby, G., Phillips, D.H., Sivakumar, V., Pakys, A., 2014. Landfill cap models under simulated climate change precipitation: impacts of cracks and root growth. *Géotechnique* 64 (2), 95–107.
- Snow, D.T., 1969. Anisotropic permeability of fractured media. *Water Resour. Res.* 5, 1273–1289.
- Spyreas, G., Gibson, D.J., Middleton, B.A., 2001. Effects of endophyte infection in tall fescue (*Festuca arundinacea*: Poaceae) on community diversity. *Int. J. Plant Sci.* 162, 1237–1245.
- Stuedemann, J.A., Hoveland, C.S., 1988. Fescue endophyte: history and impact on animal agriculture. *J. Prod. Agric.* 1, 39–44.
- Waldron, L.J., 1977. The shear resistance of root-permeated homogeneous and stratified soil. *Soil Sci. Soc. Am. J.* 41 (5), 843–849.
- Waldron, L.J., Dakessian, S., 1981. Soil reinforcement by roots: calculation of increased soil shear resistance from root properties. *Soil Sci.* 132 (6), 427–435.
- Wang, Q., Cui, Y.J., Tang, A.M., Barnichon, J.D., Saba, S., Ye, W.M., 2013. Hydraulic conductivity and microstructure changes of compacted bentonite/sand mixture during hydration. *Eng. Geol.* 164, 67–76.
- Waugh, W.J., Thiede, M.E., Bates, D.J., Cadwell, L.L., Gee, G.W., Kemp, C.J., 1994. Plant cover and water balance in gravel admixtures at an arid waste-burial site. *J. Environ. Qual.* 23 (4), 676–685.
- Wu, T.H., Watson, A., 1998. In-situ shear tests of soil blocks with roots. *Can. Geotech. J.* 35, 579–590.
- Yesiller, N., Miller, C.J., Inci, G., Yaldo, K., 2000. Desiccation and cracking behavior of three compacted landfill liner soils. *Eng. Geol.* 57 (1), 105–121.
- Yuen, K., Graham, J., Janzen, P., 1998. Weathering-induced fissuring and hydraulic conductivity in a natural plastic clay. *Can. Geotech. J.* 35 (6), 1101–1108.
- Zhang, X., Ervin, E.H., Schmidt, R.E., 2002. Seaweed extract, humic acid, and propiconazole improve Tall Fescue sod heat tolerance and posttransplant quality. *Hortic. Sci.* 38 (3), 440–443.
- Zhou, Y., Chen, J., Wang, X., 2009. Research on resistance cracking and enhancement mechanism of plant root in slope protection by vegetation. *J. Wuhan Univ. (Nat. Sci. Ed.)* 55 (5), 613–618 (in Chinese).
- Zhu, H., Zhang, L.M., 2015. Evaluating suction profile in a vegetated slope considering uncertainty in transpiration. *Comput. Geotech.* 63, 112–120.